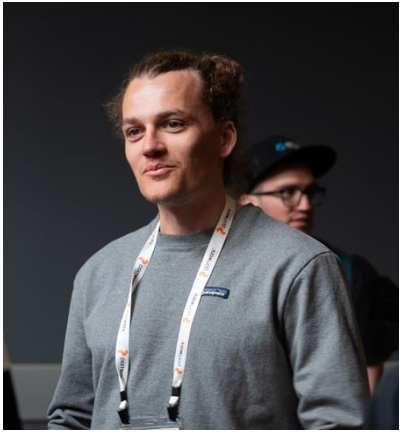
swisscom

The Accidental Platform Team: Kubernetes Operators at Swisscom

March 2026, Jelena Malić, Fabian Schulz



Fabian Schulz

DevOps Engineer

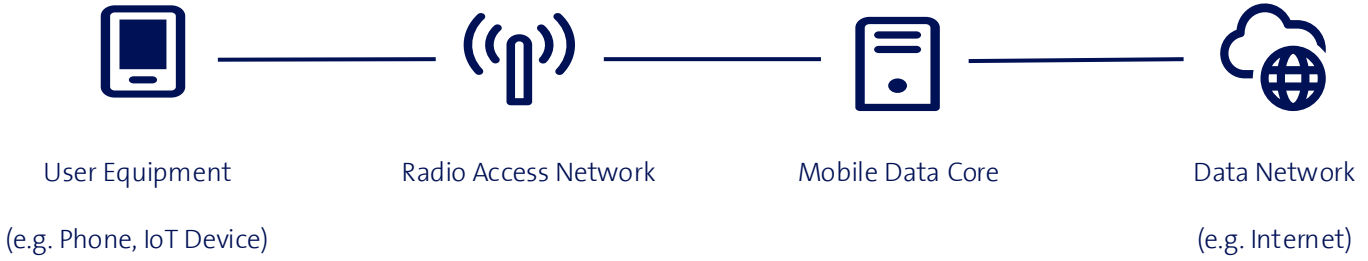
fabian.schulz1@swisscom.com

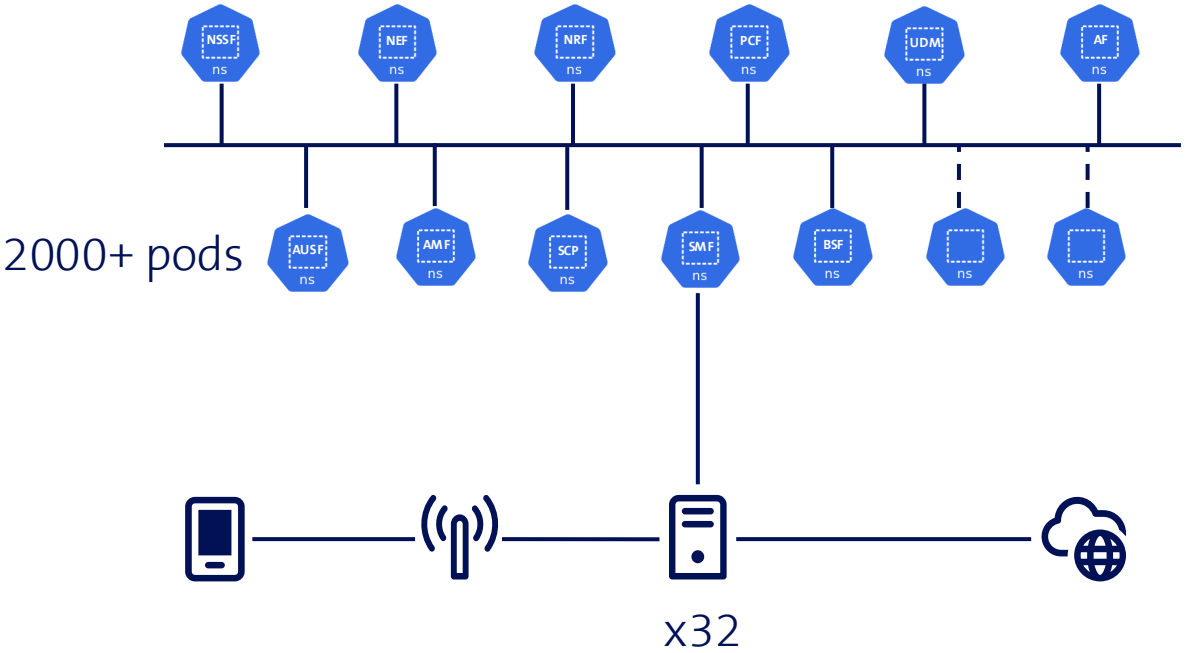
Jelena Malić

Product Owner

jelena.malic@swisscom.com

swisscom

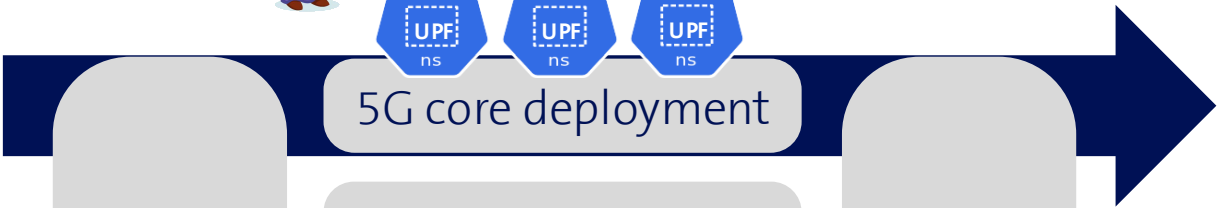

User Equipment
(e.g. Phone, IoT Device)
Radio Access Network
Mobile Data Core
Data Network
(e.g. Internet)


NSSF
ns
NEF
ns
NRF
ns
PCF
ns
UDM
ns
AF
ns
2000+ pods
AUSF
ns
AMF
ns
SCP
ns
SMF
ns
BSF
ns
ns
ns
x32

DC
DC
DC

Network Setup Wizard
Welcome to the Network Setup Wizard
This wizard will help you set up this computer to run on your network. With a network you can:
• Share an Internet connection
• Set up Internet Connection Firewall
• Share files and folders
• Share a printer
To continue, click Next.
< Back
Next >
Cancel

Yer a wizard, Harry.



routers

Kubernetes

OS

Server Hardware

routers

Create IP assignments

2 days

Collect IP in IPAM

2 days

Populate inventory

1 day

Create change requests

5-10 days

MetalLB

k8s /CNF

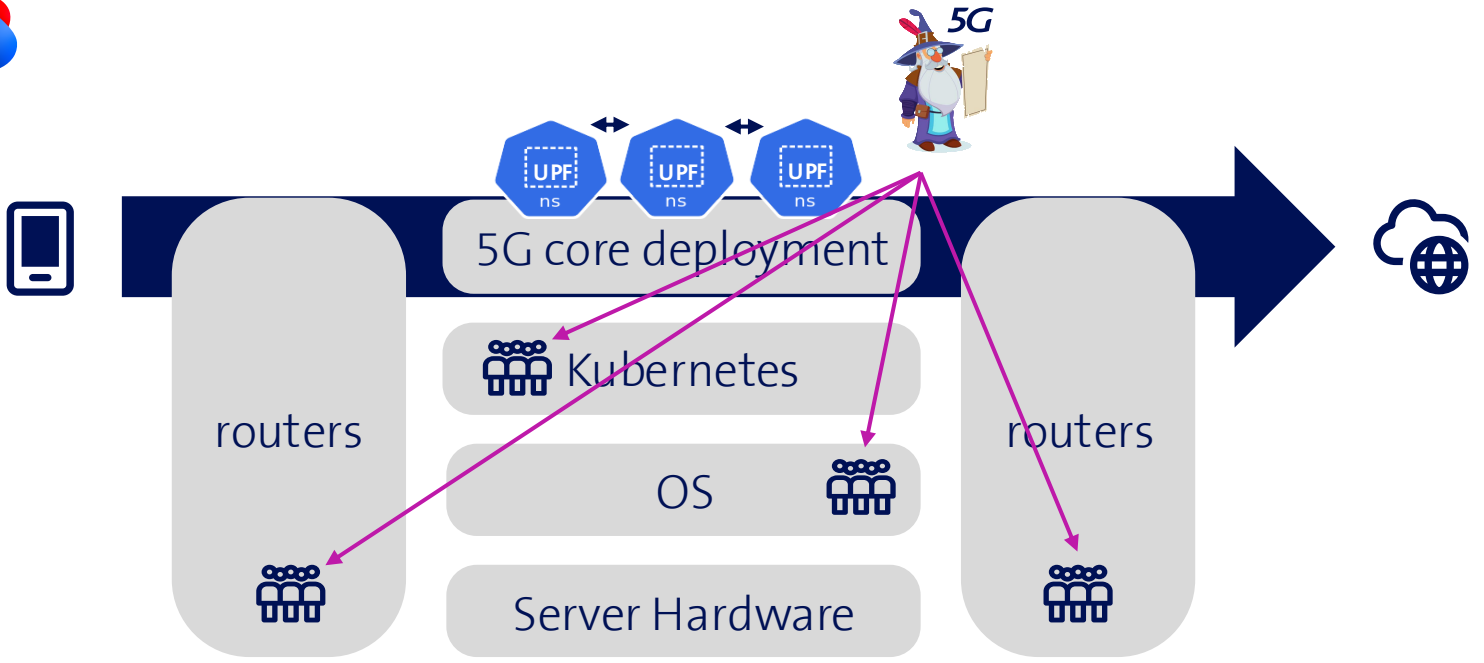
Thor UI

edge router

Integrate, test and observe

5 - 10 days

Landslide


5G
UPF
ns
UPF
ns
UPF
ns
5G core deployment
Kubernetes
OS
Server Hardware
routers
routers

Easy, right?

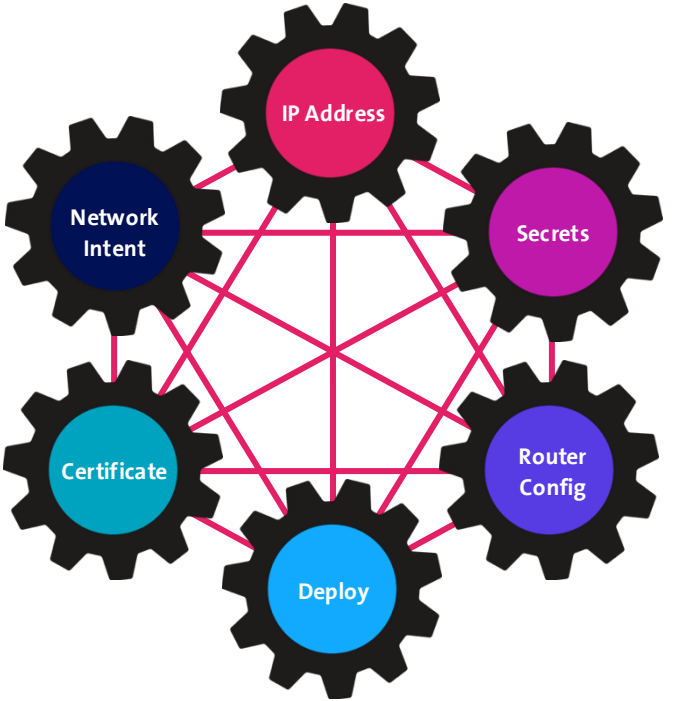
Pain Points When Deploying a 5G Core Network

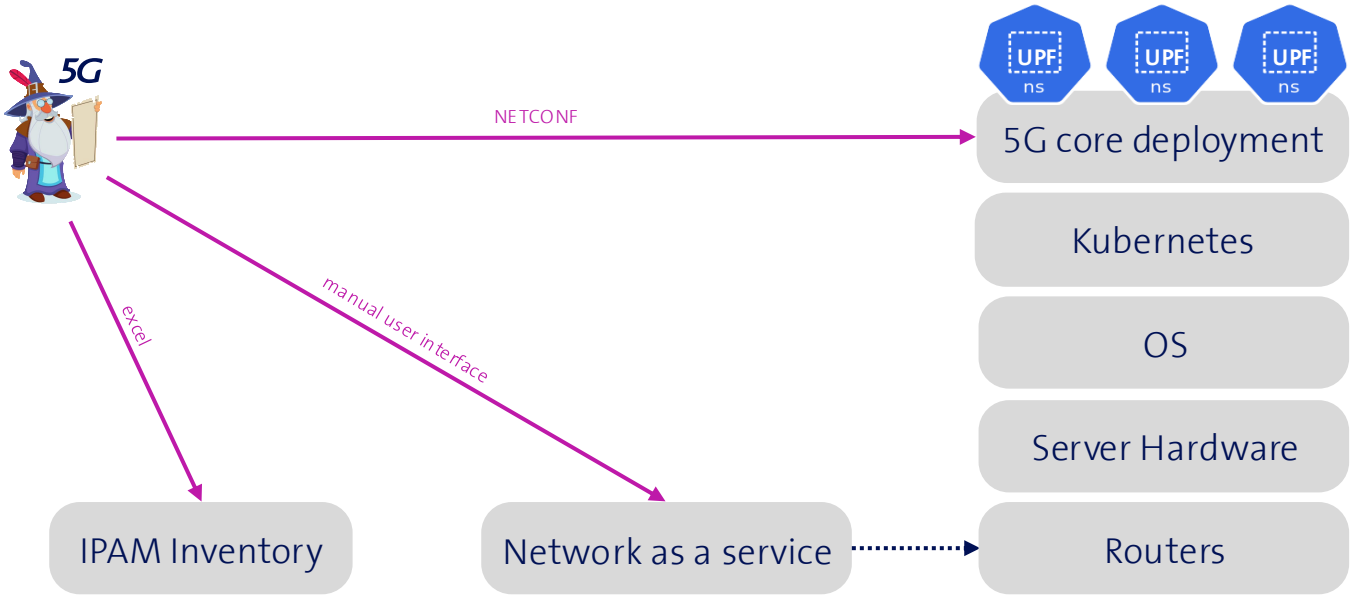
Cross-domain Knowledge

Manual Process

Static Config

high
level
network
intent
5G


IP Address
Network Intent
Secrets
Certificate
Router Config
Deploy


5G
NETCONF
UPF
ns
UPF
ns
UPF
ns
5G core deployment
Kubernetes
OS
Server Hardware
Routers
excel
manual user interface
IPAM Inventory
Network as a service

high level
intent YAML

K8s
API

Kubernetes
Operators

5G core deployment

Kubernetes

OS

Server Hardware

Routers

IPAM Inventory

Network as a service

Cross-domain Knowledge

Domain Knowledge in Operators

Manual Process

Automation

Static Config

Constant Reconciliation

More details?

https://architecture.cncf.io/architectures/swisscom-cloud-native-telco

How brave do you feel?

high level
intent YAML

K8s
API

Automation
Potions

5G core deployment

Kubernetes

OS

Server Hardware

Routers

Network as a service

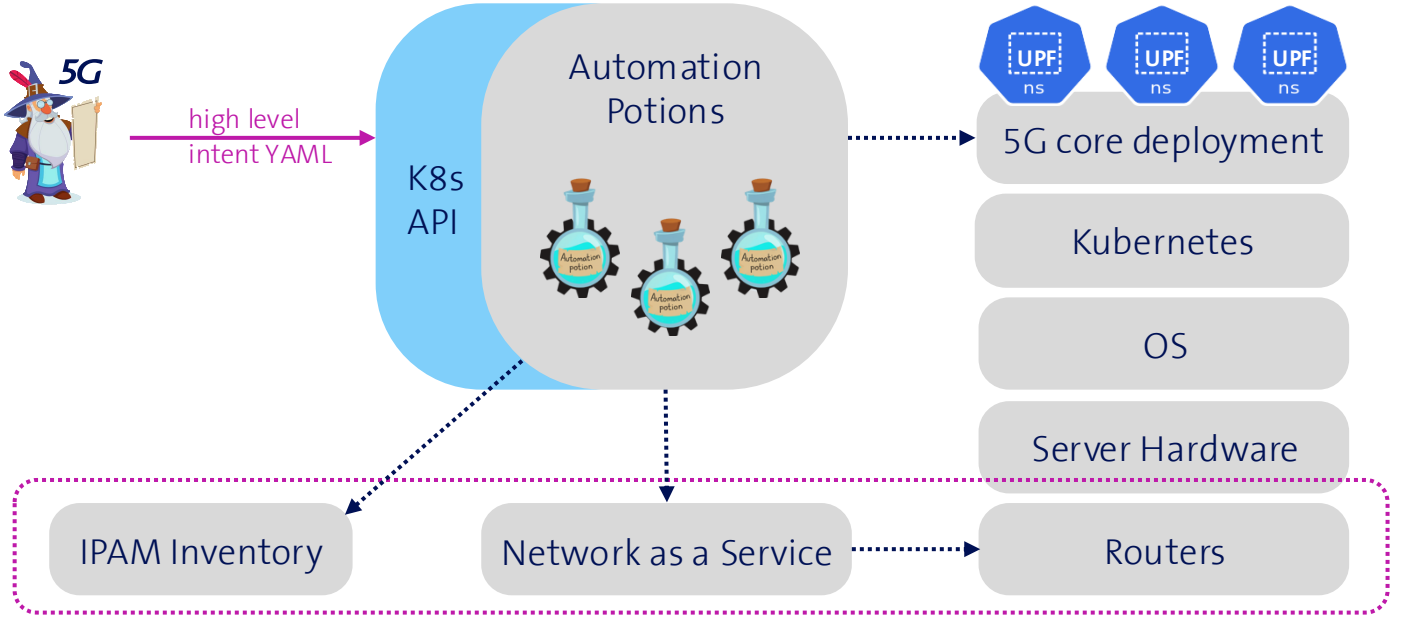
IPAM Inventory

Gradual Approach

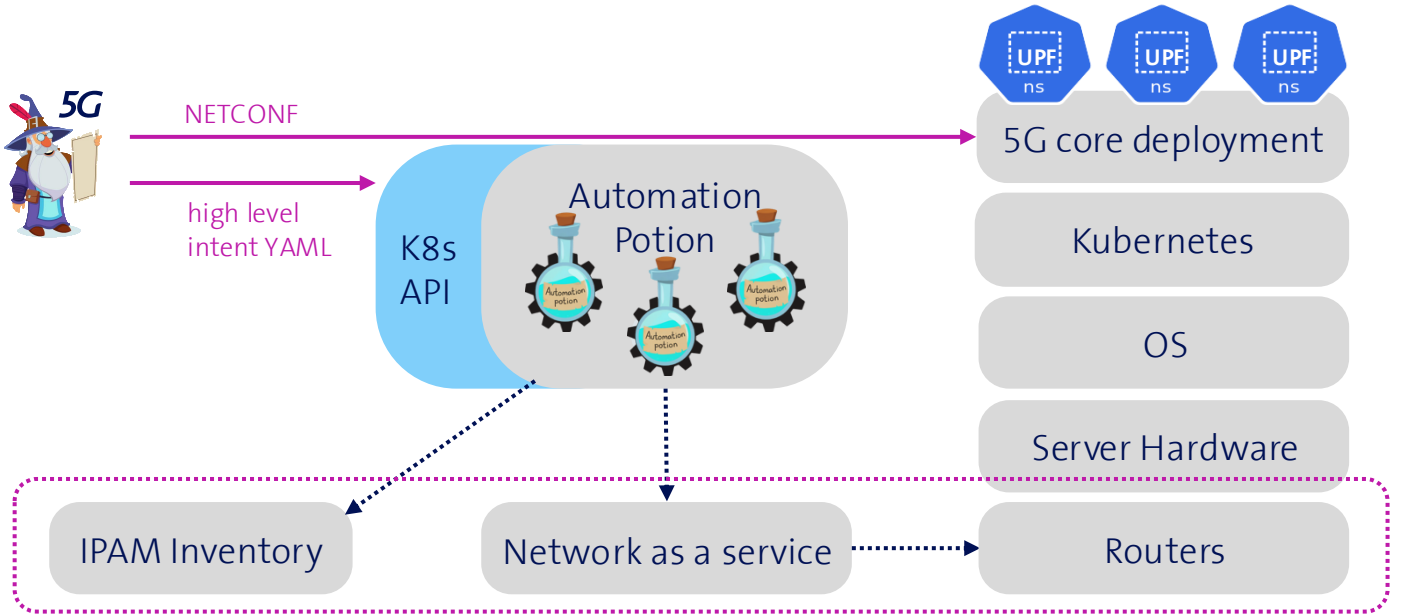
Select first easy use case

Ship first prototype early

Iterate on feedback

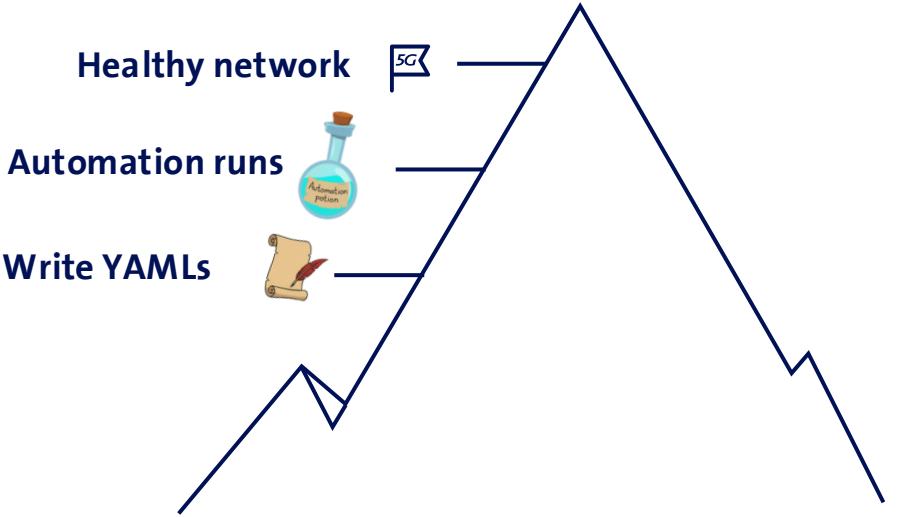


First Use Case



First Use Case

5G


Healthy network
5G
Automation runs
Automation potion
Write YAMLs

Let’s goooo

```go
// CcdSecondaryNetworkTopologySpec defines the desired state of CcdSecondaryNetworkT
// +kubebuilder:validation:XValidation:rule="(!self.cpConfig.topology.contains("Bgp",
type CcdSecondaryNetworkTopologySpec struct {
        // Configuration needed for IPAM in Netbox
        NetBoxConfig NetBoxConfigTopology `json:"netBoxConfig,omitempty"`

        // Configuration needed for management of ConnectionPoints in CPBroker
        CpConfig CpConfig `json:"cpConfig,omitempty"`

        // InternalNetwork is used to determine if the network is internal or extern
        // If set to true, the CP will be created with createRe
        // Default: false
        // Immutable after creation.
        // +default:value=false
        // +kubebuilder:validation:Optional
        // +kubebuilder:validation:XValidation:rule="self == ol
        InternalNetwork bool `json:"internalNetwork,omitempty"`

        // Interval for MetalLB BFDProfile receiveInterval, Met
        // Must be between 10 and 3000
        // Default: 300
        // +default:value=300
        // +kubebuilder:validation:Minimum=10
        // +kubebuilder:validation:Maximum=3000
        BfdInterval int32 `json:"bfdInterval,omitempty"`

        // Determines the number of consecutive missed BFD pack
        // Must be between 2 and 10
        // Default: 4
        // +default:value=4
        // +kubebuilder:validation:Minimum=2
        // +kubebuilder:validation:Maximum=10
        BfdMultiplier int32 `json:"bfdMultiplier,omitempty"`

        // Defines the ASN used on the CNF side.
        // This field is required and immutable once set
        // Required
        // Example: "65002"
        // +kubebuilder:validation:XValidation:rule="self == oldSelf",message="Field
        CnfAsn string `json:"cnfAsn,omitempty"`

        // Configuration of the git repository for the fabric information export
        // Default: {}
        // +kubebuilder:default:={}
        FabricOutputConfig FabricOutputConfig `json:"fabricOutputConfig"`
}
```

```go
// +kubebuilder:validation:XValidation:rule="(self.ipv6.enable || self.ipv4.enable)"
type NetBoxConfigTopology struct {
        // NetBox Tenant that will be used for creating NetBox IPAM Resources
        // Example: "5GDMC"
        // Required
        // Immutable after creation.
        // +kubebuilder:validation:Required
        // +kubebuilder:validation:XValidation:rule="self == oldSelf",message="Field 'tenant' is immutable"
        Tenant string `json:"tenant,omitempty"`
        // Indicating if we are selecting IPv6 IPs, and the selectors used
        // Required

                                                                    quired
                                                                    lidation:rule="self == oldSelf",message="Field 'ipv6' is immutable"
                                                                     `json:"ipv6,omitempty"`
                                                                    ting IPv6 IPs, and the selectors used

                                                                    quired
                                                                    lidation:rule="self == oldSelf",message="Field 'ipv4' is immutable"
                                                                     `json:"ipv4,omitempty"`

                                                                    IPv6 selection criteria.
                                                                    ule="(self.enable == false || (has(self.parentPrefixSelector) && has(self.pref

                                                                    ned

                                                                    in NetBox Operator. Note that the "family" field will be added by the Controll
                                                                     family (IPv4/IPv6)

                                                                    dation:rule="self == oldSelf",message="Field 'parentPrefixSelector' is immutab
        parentPrefixSelector map[string]string `json:"parentPrefixSelector,omitempty"`

        // The desired prefix length for the IPv6 prefix. Be aware that this is an integer which will be converted to
        // string in slash notation for NetBox Operator
        // Immutable after creation.
        // Must be between 8 and 128
        // Example: 64
        // +kubebuilder:validation:Minimum=8
        // +kubebuilder:validation:Maximum=128
        // +kubebuilder:validation:XValidation:rule="self == oldSelf",message="Field 'prefixLength' is immutable"
        PrefixLength int `json:"prefixLength,omitempty"`
}
```

We are done, right?

Maybe not quite:

- We don’t have time to test your stuff.
- You cannot deploy a 5G Core with GitOps because there will always be manual changes on the box.
- How can I see which changes will be applied to the cluster?
- We had a UI which was more intuitive!
- How can I be sure it worked?
- I think we were faster the old way.
- How do we know where this change is coming from?
- I don’t understand Kubernetes.
- I don’t even know how to start using it.
- I just did it myself using a Gitlab pipeline, it’s easier that way.

Oops we just built a platform...

- We don’t have time to test your stuff.
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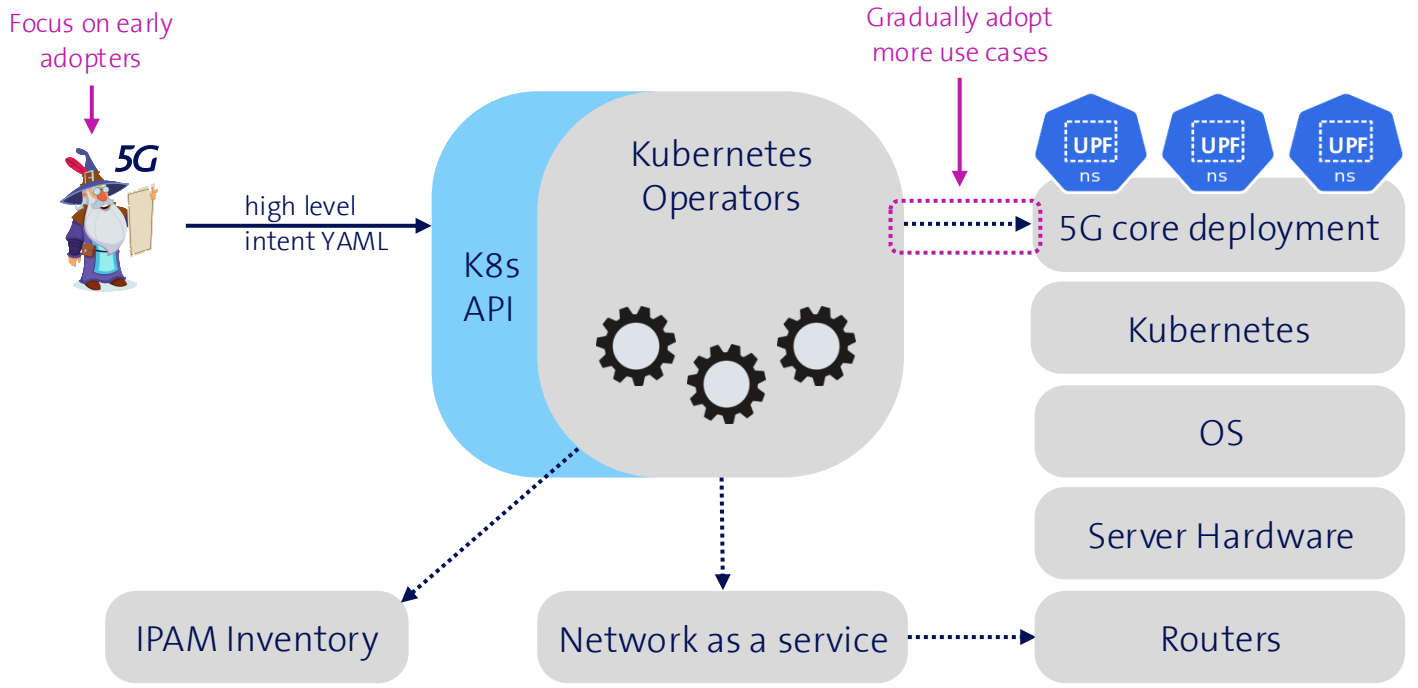
Gradual Adoption

How do we get our solution tested?



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Graphical UI

CR(D) Wizard

Filter CRDs...

BFDProfile
bfdprofiles.metallb.io
1 in

BGPAdvertisement
bgpadvertisements.metallb.io
1 in

BGPPeer
bgppeers.metallb.io
2 in

CcdPrimaryNetworkTopology
ccdprimarynetworktopologies.configassembly.swisscom.com
1 in

CcdSecondaryNetworkTopology
ccdsecondarynetworktopologies.configassembly.swisscom.com
Not in

Community
communities.metallb.io
Not in

Schema Definition (v1beta2)

`apiVersion` string

APIVersion defines the versioned schema of this representation of an object. Servers should convert recognized schemas to the latest internal value, and may reject unrecognized values.
More info: https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources

`kind` string

Kind is a string value representing the REST resource this object represents. Servers may infer this from the endpoint the client submits requests to. Cannot be updated.
In CamelCase.
More info: https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds

`metadata` object

`spec` object

BGPPeerSpec defines the desired state of Peer.

`bfdProfile` string

The name of the BFD Profile to be used for the BFD session associated to the BGP session. If not set, the BFD session won't be set up.

`connectTime` string

Requested BGP connect time, controls how long BGP waits between connection attempts to a neighbor.

Headlamp

Search Press / to search

Hello

- Home
- Clusters
  - Namespaces
  - Nodes
  - Advanced Search (Beta)
- Map
- Workloads
- Storage
- Network
- Gateway (beta)
- Security
- Configuration
- Custom Resources
- Flux

+ Create

Overview

CPU
2 units

Memory
7.74 GB

Pods
18 / 19 Requested
94.7 %

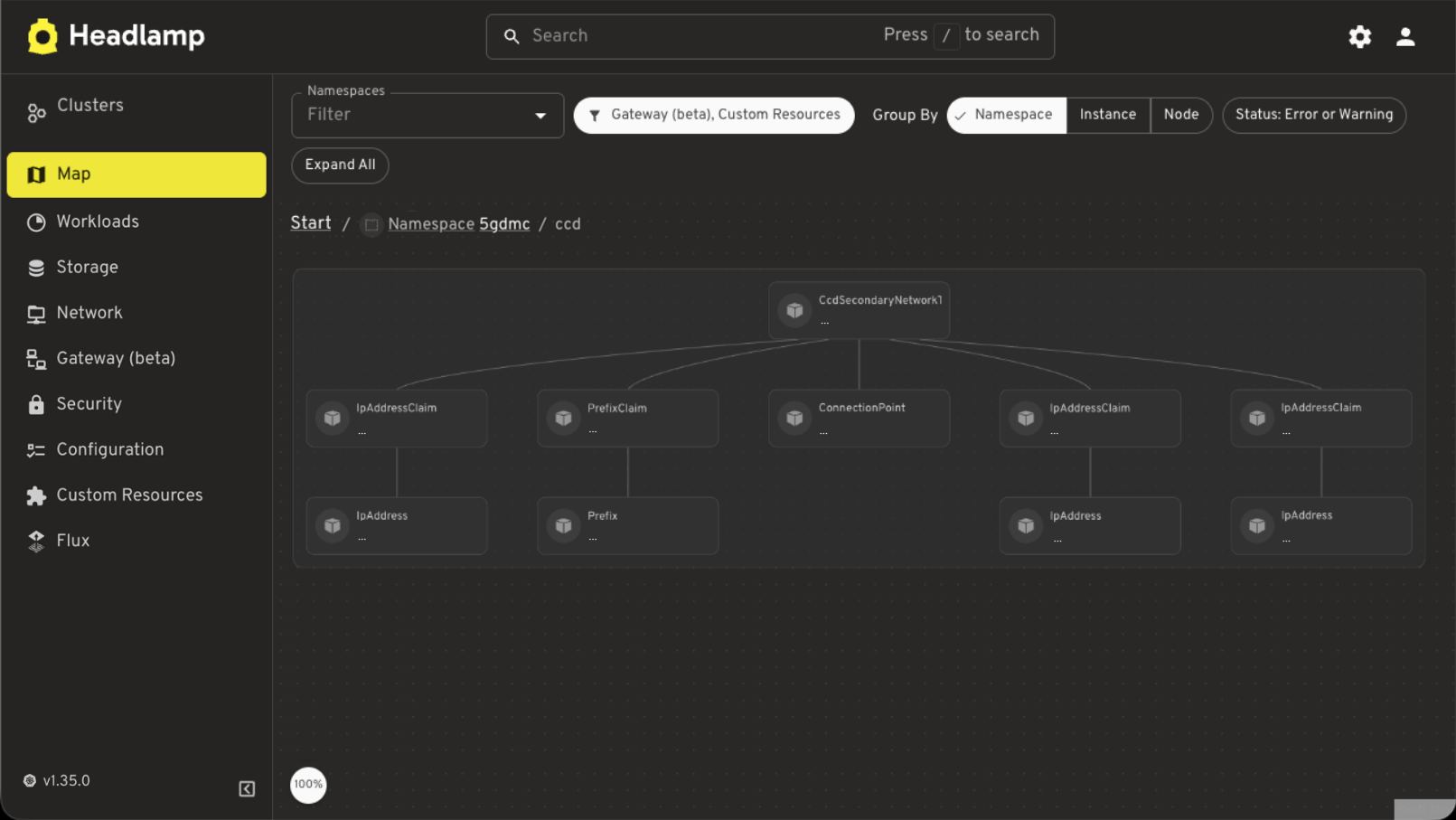
Nodes
1 / 1 Ready
100.0 %

Events

Only warnings (18)

Namespaces
Filter

Type	Name	Namespace	Reason	Message	Last Seen	Actions
Pod	netbox	default	FailedScheduling	0/1 nodes are available: 1 Insufficient cpu. no new claims to deallocate, preemption: 0/1 nodes are available: 1 No preemption victims found for incoming pod.	3m	
Pod	crd-wizard	default	BackOff	Back-off restarting failed container crd-wizard in pod crd-wizard	34m	
Pod	crd-wizard	default	FailedScheduling	0/1 nodes are available: 1 Insufficient cpu. no new claims to deallocate, preemption: 0/1 nodes are available: 1 No preemption victims found for incoming pod.	36m	


Headlamp
Search
Press / to search
Clusters
Map
Workloads
Storage
Network
Gateway (beta)
Security
Configuration
Custom Resources
Flux
v1.35.0
Namespaces
Filter
Gateway (beta), Custom Resources
Group By
Namespace
Instance
Node
Status: Error or Warning
Expand All
Start / Namespace 5gdmc / ccd
CcdSecondaryNetwork1
IpAddressClaim
PrefixClaim
ConnectionPoint
IpAddressClaim
IpAddressClaim
IpAddress
Prefix
IpAddress
IpAddress
100%

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- How can I be sure it worked?
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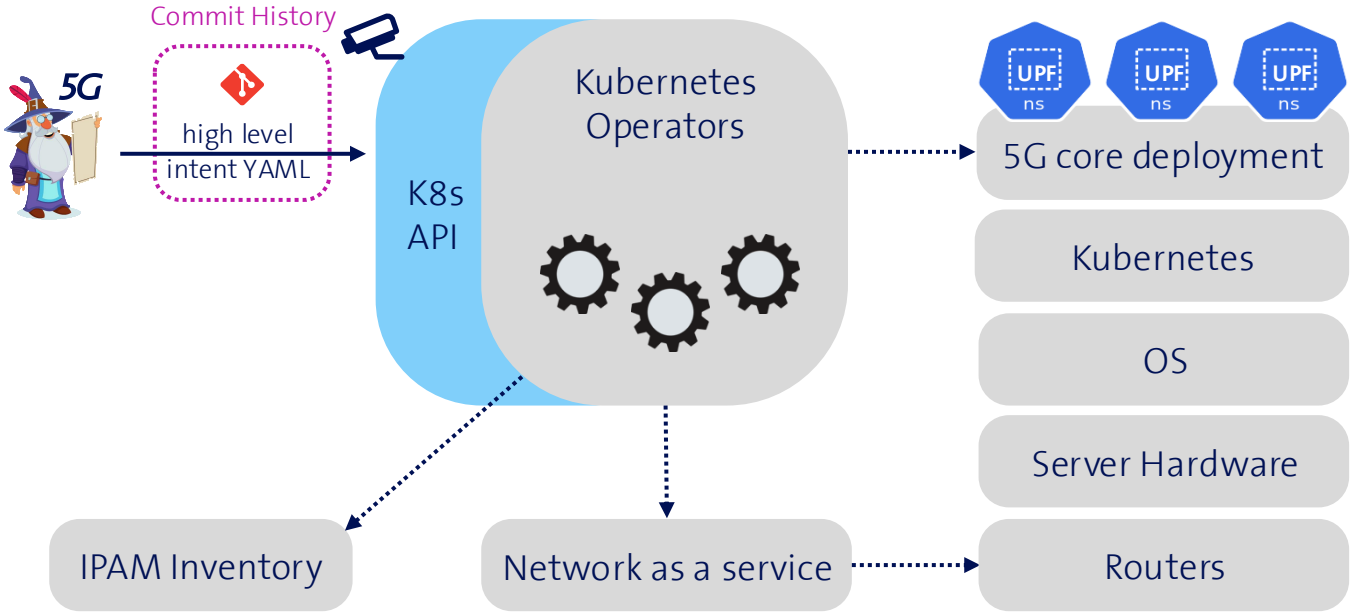
Observability and Traceability

How do we know what happened?



Tenants

5G

Total Connection Points	Total Static Routes	Prefix Claims	IP Range Claims
77	144	491	134

Primary Networks	Secondary Networks	IP Address Claims	MetalLB IP Pools
32	27	205	169

Network Instances	Network Instances
6	6

Network Instance

Network Instance	Status
	Ready
	Ready
	Ready
	Ready

Network Instance

Network Instance	Status
	Ready
	Ready
	Ready
	Ready

Primary Networks

Primary Network Na	Status
	Ready
	Ready
	Ready
	Ready

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- How do we know where this change is coming from?

> **Observability and Traceability**

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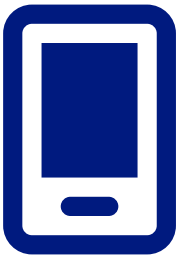
Knowledge Sharing

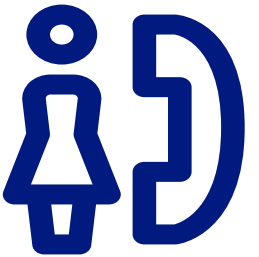
Establish Trust with Users

Design / Feedback Sessions

Focused Communication Channels



Regular Check-ins



Knowledge-Sharing Initiatives

- We don’t have time to test your stuff.
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> **Gradual Adoption**

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> **Observability and Traceability**

- I don’t understand Kubernetes.
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> **Knowledge Sharing**

The Tale of the Accidental Platform Team

Does it actually work?

Create IP tables

2 days

Create high-level intent

2 hours

Collect IP in IPAM

2 days

Populate inventory

1 day

Create lower-level intents

5 seconds

Create change requests

5-10 days

Realize intents / create CRs

5-10 minutes

Integrate, test and observe

5 - 10 days

Integrate, test and observe

2 hours and continuous testing & monitoring

And how brave do you feel now?

Thanks to colleagues for what we've achieved **together** in the past **1.5 years**.

We just created all endpoints **without a single error!**

I'm convinced that we've **saved a week or two of toil**.

Now, it took me **less than one hour** to create those networks!

What does the future hold?

Automation of all steps

What do we do regarding manual steps?

Different types of systems

How about imperative systems?

AI

Where can we meaningfully use it?

How to survive as an Accidental Platform Team?

How We Are Moving from GitOps to Kubernetes Resource Model in 5G Core

KubeCon Europe

2024

Just Claim It: Simplifying Network Automation with NetBox Operator

Cloud Native Telco Day

2025

The Cloud-Native Advantage: Intent-based Network Automation (SDC)

Container Days Conference

2025

2024

Building and Operating a Highly Reliable Cloud Native DNS Service With Open-Source Technologies

Container Days Conference

2025

Keynote: Cloud Native Evolution in Telecom: 5G, 6G and Beyond!

KubeCon + CloudNativeCon Europe

Thank you!

Wizard image credits: Nikola Malić

github.com/swisscom/cloud-native-telco